TCMC · HARMONY EARS

Chord Voicing Trainer

User Manual

（和弦聲部音準練習）

App Version: v2.0

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1. Overview

“Chord Voicing Trainer” is a web-based tool from the Taiwan Choral Music Center (TCMC) that lets you type any chord symbol, freely assign each chord tone to any of 8 voice parts (S1, S2, A1, A2, T1, T2, B1, B2) at any octave, pick one part as your singing target, and play the rest as accompaniment — with real-time microphone pitch detection checking whether you're matching the target.

Unlike the companion apps “Harmonic Third Pitch Trainer” and “Harmonic Seventh Pitch Trainer,” this tool uses standard 12-tone equal temperament rather than Just Intonation. It's built for practicing arbitrary chord voicings and pitch recognition, not for training the blended, beat-free sound of Just-Intonation intervals.

Key Features

- Type any chord symbol (e.g. Am7(-5), G7, Cmaj7, CmM7, C7(b9,#11)) and have all its chord tones parsed automatically
- Freely assign any chord tone at any octave (within A2–A5) to any of the 8 voice parts (S1/S2/A1/A2/T1/T2/B1/B2)
- Piano-keyboard note picker: tap a voice part, then tap a chord-tone button to reveal it on the keyboard — no note-name codes to read
- Each voice part has its own play/mute toggle, so you can preview the voicing while still configuring it
- Pick one part as the singing target; freely toggle the rest on or off as accompaniment — the target can be changed with one tap at any time
- Real-time microphone pitch detection with a ±6 cent tolerance band, shown live on the chart
- Chinese / English interface toggle

> 💡 *Tip: headphones are recommended. Over speakers, the accompaniment can bleed into the mic and throw off the pitch judgement.*

2. Interface Overview

Chord Input & Chord-Tone Buttons

Type a chord symbol into the “Chord Input” field and press “Parse” (or just hit Enter). The app determines the root and chord quality, lists every chord tone, and labels each one with a numbered-notation scale degree (always relative to C major).

Below the parsed result, each chord tone appears as its own button (e.g. “🔊 6·A”). These buttons do two things: tap one to hear that pitch immediately, and — during voice-part assignment — the same buttons also act as the keyboard's “reveal” switch. See “Voice Part Assignment” below.

Voice Part Assignment: Pick a Part, Then Reveal the Keyboard

Once a chord parses successfully, the 8 voice parts appear (S1, S2, A1, A2, T1, T2, B1, B2 — Soprano 1/2, Alto 1/2, Tenor 1/2, Bass 1/2) along with a piano keyboard. The keyboard starts completely dark, identical to a plain keyboard — it won't tell you upfront which keys are chord tones. This is deliberate: it avoids nudging you toward a specific key before you've decided what you're assigning.

To assign a part: tap a voice part first (e.g. B1), then tap the matching chord-tone button above (e.g. "🔊 6·A") — every not-yet-used octave of that tone across the whole range lights up. Tap one of the lit keys to complete the assignment. The keyboard then shows the part's label directly on that key, the other lit octaves of the same tone go dark again, and the chord-tone button returns to normal.

- An already-assigned key always stays lit and labeled, whether or not it's currently "revealed" — so you can see your whole voicing at a glance
- To have two parts sing the same chord tone an octave apart (octave doubling), just tap the same chord-tone button again — the still-unused octaves light back up for you to pick from
- Tap an already-assigned part to move it to a different key; use the "×" next to it to remove the assignment
- Every key and every part can be tapped to preview its sound (no need to arm a part first) — handy for exploring what each pitch actually sounds like

> 💡 *The first voice part you assign automatically becomes the singing target; you can change this anytime (see below).*

Singing Target & Playback Controls

Once parts are assigned, each part's label shows a 🎯 button — tap it to make that part your singing target (highlighted in gold); every other row then shows its own independent play/mute button. An "Assigned Parts & Singing Target" list also appears below, showing each part's label, note name, scale degree, and frequency, where you can switch the target the same way.

- ▶ Play Chord: plays every assigned part together (including the target) — good for hearing the full configured harmony at once
- ▶ Play Target: plays only the target part, so you can re-check the pitch as many times as you like; press again to stop
- The Play/Mute button next to each non-target part independently toggles that part's accompaniment playback
- ▶ Start: begins the practice sequence — briefly demos the target note, then plays whichever accompaniment parts you've enabled, and starts microphone detection
- ■ Stop: ends the session, stopping all playback and mic detection
- Enable/Disable Mic: manually control whether your voice is being detected

3. How to Use

Click here to open: Chord Voicing Trainer (GitHub Pages)

1. Type the chord you want to practice into “Chord Input” and press “Parse.”
2. Check the parsed chord tones — tap the chord-tone buttons to preview each one.
3. Tap a voice part (e.g. start with B1/Bass, then work upward through the other parts).
4. Tap the matching chord-tone button — every not-yet-used octave of that tone lights up.
5. Tap one of the lit keys to complete the assignment. The first part you assign automatically becomes the singing target.
6. Repeat steps 3–5 for the other parts you need; tap the same chord-tone button again for octave doubling.
7. Once configured, tap the 🎯 next to any part anytime to change the singing target.
8. Toggle other parts' playback as needed, or press “▶ Play Chord” to hear the full harmony once.
9. Press “▶ Start” — the app demos the target note, then plays your enabled accompaniment parts and turns on the mic.
10. Sing the target part, aligning with the gold target line on the chart (±6 cent tolerance).
11. When you're done, press “■ Stop,” or just change the chord/voicing to start the next round.

> 💡 *The first time you use the app, your browser will prompt for microphone permission — choose Allow, or real-time pitch detection won't work.*
>
> 💡 *If you're using an in-chat preview window, security restrictions may block the permission request — download the HTML file and open it directly in a browser instead.*

4. Chord Symbols & Alteration Modifiers

Common Chord Symbols

Input	Chord Quality	Parsed Tones
C	Major triad	C, E, G
Cm	Minor triad	C, Eb, G
C7	Dominant 7th	C, E, G, Bb
Cmaj7	Major 7th	C, E, G, B
Cm7	Minor 7th	C, Eb, G, Bb
CmM7	Minor-major 7th (mMaj7)	C, Eb, G, B
Cdim7	Diminished 7th	C, Eb, Gb, A

Input	Chord Quality	Parsed Tones
Cm7(-5)	Half-diminished (m7b5)	C, Eb, Gb, Bb
C9	Dominant 9th	C, E, G, Bb, D
Csus4	Suspended 4th	C, F, G

> 💡 *Minor-major 7th chords can also be written as Cm(maj7) or CmMaj7 — all three spellings, including CmM7, parse to exactly the same tones.*

Alteration Modifiers

Add parentheses after a chord symbol to alter or add extension tones — flat 5th, sharp 11th, flat 13th, and so on. The rule is simple:

- If the chord already has a tone at that scale-degree slot (e.g. the 5th), the modifier replaces it
- If the chord has no tone at that slot (e.g. an 11th or 13th), the modifier adds a new extension tone
- Multiple modifiers can be combined, comma-separated, e.g. (b9,#11)
- + and # both mean sharp and behave identically; - and b both mean flat and behave identically

Input	What happens	Parsed Tones
Cm7(-5)	Replaces the existing 5th with a flat 5th	C, Eb, Gb, Bb
Cm7(#11)	No 11th in the base chord, so a sharp 11th is added	C, Eb, G, Bb, F#
Cm7(-13)	No 13th in the base chord, so a flat 13th is added	C, Eb, G, Bb, Ab
C7(b9,#11)	Two alterations applied at once, comma-separated	C, E, G, Bb, Db, F#

> 💡 *Parsing a new chord resets any voice-part assignments you've already made — you'll need to reassign them. Supported chord types cover common triads, 6th, 7th, 9th, and suspended chords, minor-major 7th/9th chords, plus bracketed alteration modifiers; unusual or highly specialized jazz chord symbols may not be recognized — the app will show a clear error message in that case.*

5. Pitch Chart & Tuning Judgement

The chart's vertical range adjusts dynamically to fit whichever parts are currently assigned. Thick lines mark each assigned part's pitch; gold marks the current singing target.

Your live pitch trace appears in green when within ±6 cents of the target, and red when outside that band — so you can judge intonation by color alone. Below the chart you'll also see the detected frequency, the offset in cents from the target, a text judgement (in tune / sharp / flat), and signal diagnostics (RMS, clarity) for more advanced reference.

> 💡 *The tolerance is fixed at ±6 cents (about 6% of an equal-tempered semitone) — generally the range within which the ear can clearly notice a pitch discrepancy.*
>
> 💡 *The microphone's pitch-detection range automatically adapts to cover every currently assigned part's register, not just the target's — so whichever part you're actually singing, target or not, gets detected correctly, without mistaking it for an octave jump just because it's far from the target (e.g. testing a low Bass note while the target is a high Soprano).*

6. Troubleshooting

Microphone not responding / can't detect pitch

- Confirm your browser has allowed this page to use the microphone (check the permission icon in the address bar)
- If you're using an in-app or in-chat preview window, security restrictions may block the microphone — download the HTML file and open it directly in a browser instead
- Confirm the URL starts with HTTPS; a plain local file (file://) often can't get microphone access on some mobile browsers

Tapped a chord-tone button but nothing lights up on the keyboard

- You need to tap a voice part first (S1–B2) — tapping a chord-tone button before that only plays a preview sound, it won't reveal any keys
- Already-assigned octaves of a tone stay lit without needing to be revealed again; for octave doubling, remember to tap a voice part first, then the chord-tone button again

Play Chord / Play Target has no sound

- Check that your device isn't muted and the browser tab hasn't been silenced by the system
- Some mobile browsers need one manual tap before they'll unlock audio playback at all — if there's no sound the first time, try tapping again

Pitch line jumps around or the judgement feels unstable while singing

- Headphones are recommended over speakers, to avoid the accompaniment bleeding into the mic and throwing off detection
- Sing in a quiet environment and keep a consistent distance from the microphone

Chord symbol won't parse

- Confirm it starts with A–G (optionally followed by # or b), e.g. "Bb7," "F#m7"
- Supported chord types cover common triads, 6th, 7th, 9th, and suspended chords, minor-major 7th/9th chords, plus bracketed alteration modifiers;

unusual or highly specialized jazz chord symbols may not be recognized — the app will show a clear error message in that case

7. Version History

Current version: v2.0

Development Highlights

- v1.0: initial release, with voice-part pitches assigned via dropdown menus (e.g. picking “C3” or “Eb4” from a list)
- v2.0: the voice-part assignment interface was fully redesigned as a tappable piano keyboard, with a two-step “pick a part, then reveal the tone” flow (the keyboard starts fully dark to avoid misleading the user), a one-tap 🎯 target-switch button next to each part, and the part palette moved above the keyboard; also fixed a bug where the microphone's pitch-detection range was locked to only the target note, causing octave-detection errors when testing other assigned parts; and expanded chord parsing to support minor-major 7th/9th chords (e.g. CmM7)

Production Team

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Nature of software: a web-based tool for choral voice-part rehearsal and self-practice